

Carlos Marin

Houston, TX | 713-791-3494 | carlosmarinjr1@gmail.com | [Portfolio](#) | [GitHub](#) | [LinkedIn](#)

Software Engineer | AI & Full-Stack Development

Full-stack engineer specializing in AI-powered applications — from RAG pipelines and computer vision systems to production-ready web interfaces. Proficient in Python, JavaScript (React, Node.js), and REST API development, with a focus on integrating LLMs and machine learning models into real, usable software. Dual-certified by IBM in AI Engineering and AI Development.

TECHNICAL SKILLS

- **AI & Machine Learning:** LLMs (Gemini, GPT, Llama), LangChain, RAG (Retrieval-Augmented Generation), Prompt Engineering, Computer Vision, NLP, PyTorch, TensorFlow, Scikit-learn
- **Full Stack Development:** Typescript, JavaScript (React, Node.js, Express), Python (Flask, Django, FastAPI), HTML5/CSS3, Streamlit
- **Database, DevOps & Cloud:** SQL (PostgreSQL), Docker, PySpark, Pandas, Git/GitHub, REST APIs, Vector Databases (Pinecone/Chroma), Cloud Deployment

TECHNICAL PROJECTS

Agent Data Scientist | *Python, FastAPI, Langchain, React* | [Github](#) | [Preview](#)

- Architected a full-stack AI application using FastAPI and React, engineering a LangChain-Groq pipeline to automate exploratory data analysis and Python code generation.
- Designed a server-side code execution engine using Python's `exec()`, capturing matplotlib figures as base64 images and streaming live visualizations to the React frontend
- Developed a single-service production build where FastAPI serves both the LangChain-powered REST API and compiled React frontend from one Uvicorn process, eliminating the need for a separate frontend server.

AI Cover Letter Architect | *Python, Flask, Google Gemini (LLM), Bootstrap* | [Github](#) | [Preview](#)

- Built a RAG pipeline using LangChain, Chroma, and HuggingFace embeddings to ground cover letter generation in a curated knowledge base, achieving 100% retrieval precision across ML, Full Stack, and GenAI roles.
- Engineered a custom eval framework measuring retrieval precision, grounding score, and output quality via LLM-as-judge (Gemini), delivering a repeatable scoring baseline to improve generation quality over time.
- Secured the API layer with Pydantic input validation, prompt injection sanitization, and Flask-Limiter rate limiting.

NeuroScan AI | *React, Node.js, YOLOv26, Express.js, Python* | [Github](#) | [Preview](#)

- Engineered a full-stack MRI diagnostic platform bridging Node.js and Python to classify brain scans using custom-trained YOLO computer vision model.
- Designed a child-process pipeline with custom Multer storage logic to ensure data integrity and 100% compatibility with computer vision decoders.
- Developed a React dashboard with asynchronous state management, reducing result interpretation time by 80%.

PROFESSIONAL EXPERIENCE

Houston Methodist Hospital | Houston, TX | **Clinical Research Coordinator - Heart Failure** 04/2023 - 09/2025

- Drove a 20% increase in enrollment throughput by re-engineering pre-screening workflows in collaboration with the principal investigator, identifying bottlenecks, redesigning intake logic, and reducing manual steps.
- Cut data query resolution time by 30% by designing and implementing a structured weekly data review pipeline, enabling early defect detection upstream of sponsor audits.
- Led end-to-end closeout of two Phase II studies by coordinating cross-functional action item resolution, regulatory reconciliation, and stakeholder sign-off ahead of all contractual deadlines.

EDUCATION & CERTIFICATIONS

University of Houston - Downtown - BS - Biological and Physical Sciences May 2017

IBM AI Engineer | Professional Certificate April 2026

IBM AI Developer | Professional Certificate October 2025

Codecademy | Full Stack Engineer Professional Certificate December 2025